

NETWORK FIREWALLS

The First Line of Defense in Modern Cybersecurity Architecture

TECHNICAL OVERVIEW & EVOLUTION

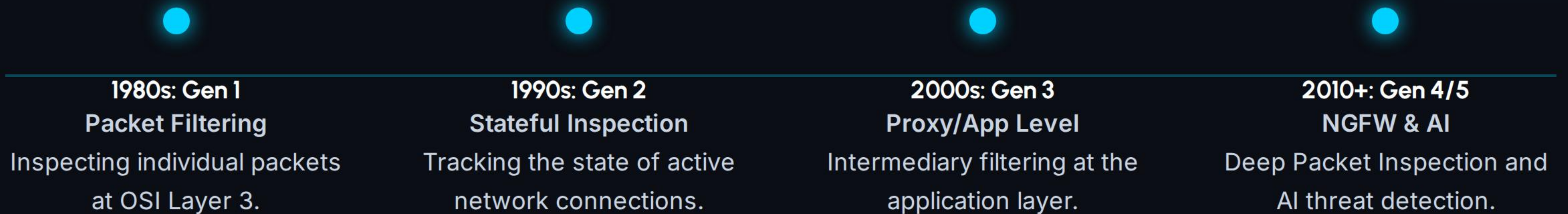
| WHAT IS A FIREWALL?

A **security system** that monitors and controls incoming and outgoing network traffic.

- 🛡️ **Digital Barrier:** Acts as a filter between a trusted internal network and an untrusted external network (Internet).
- 🛡️ **Policy Driven:** Operates based on a defined set of security rules and protocols.
- 🛡️ **Gatekeeper:** Inspects data packets to decide whether to Allow, Block, or Flag them.
- 🛡️ **Versatility:** Can be implemented as hardware, software, or cloud-based services.

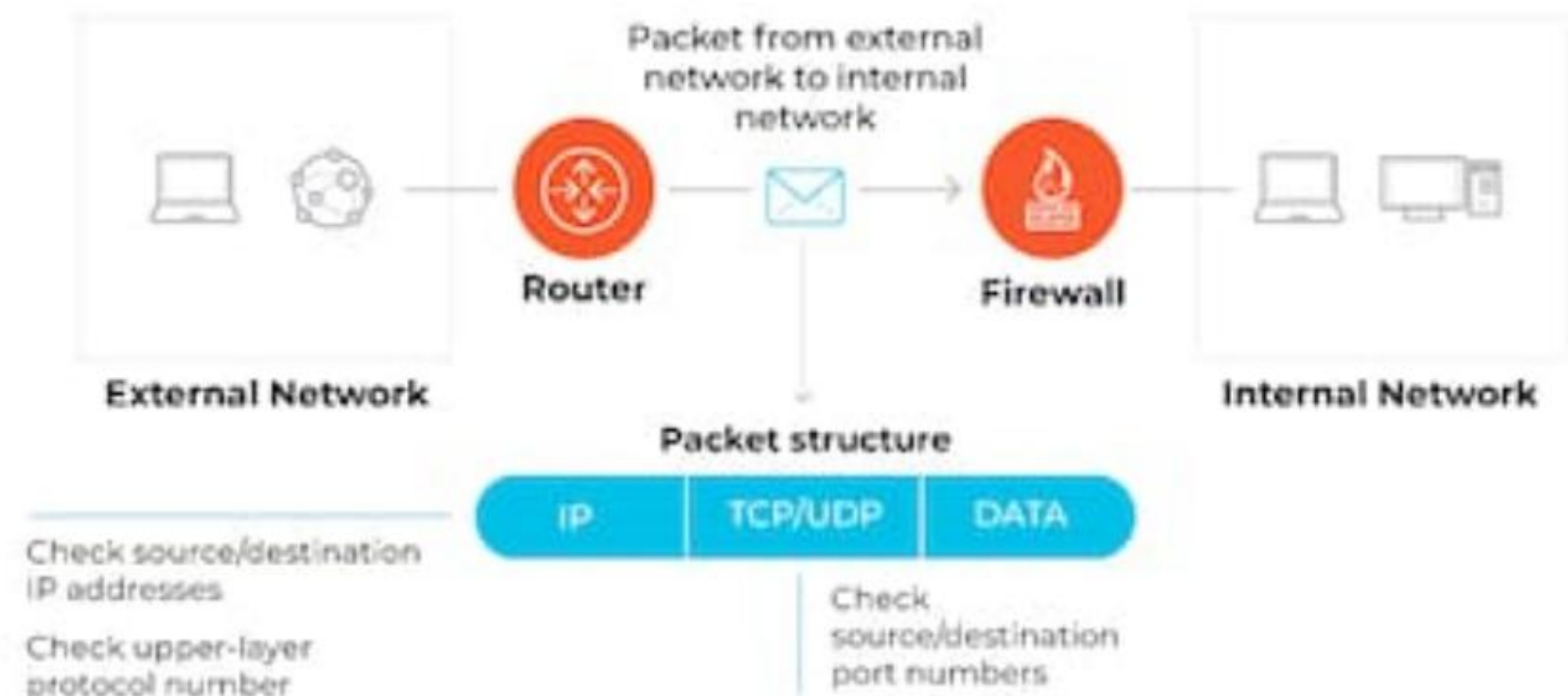


EVOLUTION OF PROTECTION



OPERATIONAL MECHANICS

How a Packet Filtering Firewall Works



The Inspection Process

Firewalls use **Rule-Based Logic** to evaluate every connection attempt.

1. Header Analysis: Checks source/destination IP, port numbers, and protocol types.

2. Policy Matching: Compares packet data against the Access Control List (ACL).

3. Payload Inspection: NGFWs look deep inside the packet to find hidden malware.

FIREWALL CLASSIFICATIONS

From Network Edge to Application Layer

PRIMARY DEPLOYMENT TYPES



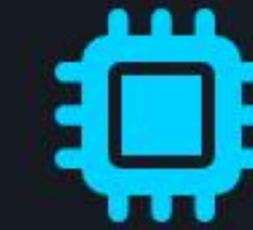
FWaaS

Firewall-as-a-Service provides cloud-native inspection for distributed teams and remote access.



WAF

Web Application Firewalls protect specific servers from Layer 7 attacks like SQL Injection and XSS.



Virtual FW

Software-defined security designed for virtualized environments and internal network segmentation.



CYBER SECURITY

— ICONS —



NEXT-GEN CAPABILITIES (NGFW)

Advanced Protection Features

- 🛡️ **Application Awareness:** Can block specific apps (e.g., Skype) while allowing others.
- 🛡️ **Integrated IPS:** Real-time intrusion prevention to stop active hacking attempts.
- 🛡️ **DPI (Deep Packet Inspection):** Scans the actual data payload for threats.
- 🛡️ **Encrypted Traffic Analysis:** Inspects SSL/TLS traffic without compromising privacy.

Identity-Based Control

Modern firewalls link traffic to specific **User Identities** rather than just IP addresses, allowing for granular "Zero Trust" policy enforcement across the enterprise.

FIREWALL COMPARISON MATRIX

Feature	Packet Filter	Stateful Inspection	NGFW (Next-Gen)
OSI Layers	Layer 3	Layer 3-4	Layer 3-7
Context Awareness	None	Connection State	App & User Identity
Security Depth	Low	Moderate	High (Deep Scan)
IPS Integration	✗	✗	✓

MARKET & SECURITY IMPACT

NGFW Market (2024)

\$13.9 Billion

Attacks Blocked/Day

Millions per Org

Remote Work Adoption

Increasing FWaaS

"Firewalls remain the #1 security investment for enterprises transitioning to hybrid cloud environments."

THE FUTURE OF FIREWALLS



AI-Driven

Automated threat hunting and predictive rule adjustments using machine learning.



Zero Trust

"Never Trust, Always Verify" integration with identity providers for micro-segmentation.



SASE Integration

Convergence of networking and security into a single cloud-delivered service model.

FLEXIBLE INFRASTRUCTURE

 Hardware and Cloud Integration

Deployment Strategy

-  **On-Premises:** High-performance hardware appliances for data centers.
-  **Cloud-Native:** Instances deployed directly in AWS, Azure, or GCP.
-  **Hybrid:** Synchronized policies between physical office and cloud resources.
-  **Endpoint:** Software firewalls running directly on host machines.

Questions?

Thank you for your time!

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 www.securityportal.com

IMAGE SOURCES



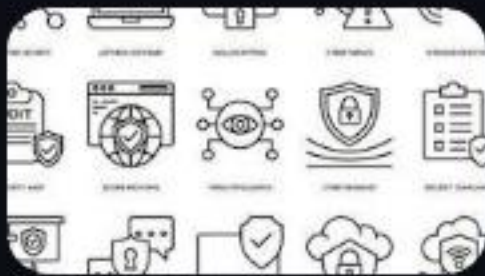
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